

Udacity - Sensor Fusion Course

Camera Based 2D Feature Tracking

Chris Holmes - Mid-Term Project Submission

Keypoints

MP.1 - Data Buffer Optimization

Implement a vector for dataBuffer objects whose size does not exceed a limit (e.g. 2 elements). This can be achieved by pushing in new elements on one end and removing elements on the other end.

Utilising the available maximum buffer size variable (dataBufferSize) and (dataBuffer) vector for current size of the buffer, a conditional statement is implemented to ensure the current size does not exceed the maximum at any given time. The statement checks the current buffer size against the preset maximum:-

```
if (dataBuffer.size() > dataBufferSize)
```

If the current size of the buffer is below or equal to the maximum, the code sequence continues. When the case does occur in which the buffer does exceed the maximum, the oldest entry is erased from the vector as the new newest entry appears. This is handled inside the if statement with the erase method.

```
dataBuffer.erase(dataBuffer.begin());
```

MP.2 Keypoint Detection

Implement detectors HARRIS, FAST, BRISK, ORB, AKAZE, and SIFT and make them selectable by setting a string accordingly.

The existing conditional if statement has been extended to further handle the HARRIS detector method and an additional conditional if statement has been implemented to handle the remaining detector methods, FAST, BRISK, ORB, AKAZE and SIFT. The string variable (detectorType) is then used for comparison to logically select the appropriate case.

```
string detectorType = "HARRIS";  
if (detectorType.compare("SHITOMASI") == 0)  
{  
    detKeypointsShiTomasi(keypoints, imgGray, false);  
}  
else if (detectorType.compare("HARRIS") == 0)  
{  
    detKeypointsHarris(keypoints, imgGray, false);  
}
```

In the example above, the HARRIS detector case would be selected due to the matching string. The instantiation of the detector is then handled in the (detKeypointsHarris) method in the (matching2D_student.ccp) file as the string is passed through.

The conditional statements for the remaining detectors are handled in the same way inside the matching2D_student file. In this case the (detectorType) string is passed through to make the string value available to the (detKeyPointsModern) method. Similarly, the instantiations for all detectors are handled inside the same file, receiving the input image and generating the detected keypoints.

MP.3 Keypoint Removal

Remove all keypoints outside of a pre-defined rectangle and only use the keypoints within the rectangle for further processing.

To handle the points only inside the defined bounded area by variable (vehicleRect) a for loop iteration function is utilised to allow the iteration through the keypoints vector. For each iteration, an if statement is then used to check if the current points exist within the bounded area or not. If there are no points inside the area, the code sequence continues. If there are, a new variable (KeeperKeyPoints) is appended with the points using the push_back method.

```
for (auto i : keypoints)  
{  
if(vehicleRect.contains(i.pt))  
{  
    KeeperKeyPoints.push_back(i);  
}  
}
```

To ensure the points are then taken forward for the rest of the code sequence, the original keypoints variable is then renamed with the new updated filtered vector.

```
keypoints = KeeperKeyPoints;
```

Descriptors

MP.4 Keypoint Descriptors

Implement descriptors BRIEF, ORB, FREAK, AKAZE and SIFT and make them selectable by setting a string accordingly.

In file (matching2D_student.ccp) the method for handling the descriptor is implemented in the function named (descKeypoints). Similarly, to how the string selection is implemented for the detectors in in MP.2, a string variable (descriptorType) is passed to (descKeypoints). This string is then evaluated using logical comparison as described in the MP.2. For the selected descriptor the opencv instantiation of the given descriptor is then applied to the variable, (extractor).

```
string detectorType = "BRIEF";
if (detectorType.compare("BRIEF") == 0)
{
    extractor = cv::xfeatures2d::BriefDescriptorExtractor::create();
}
```

MP.5 Descriptor Matching

Implement FLANN matching as well as k-nearest neighbor selection. Both methods must be selectable using the respective strings in the main function.

In file (matching2D_student.ccp) the method for handling the FLANN based matcher type is handled again using the string based logical comparison from the main file loop. The variable (matcherType) is passed from the main project file into the function (matchDescriptors) in the (matching2D_student.ccp) file. A comparison is then made to choose either the brute force matching type or FLANN matching type. The variable (matcher) is then instantiated accordingly.

In the same (matchDescriptors) function, a further string based logical comparison is made, this time with the variable (selectorType) passed in from the main file loop function. The two types of selector being, nearest neighbour (SEL_NN) and k nearest neighbour (SEL_KNN).

MP.6 Descriptor Distance Ratio

Use the K-Nearest-Neighbor matching to implement the descriptor distance ratio test, which looks at the ratio of best vs. second-best match to decide whether to keep an associated pair of keypoints.

When the appropriate string for k nearest neighbour is defined (SEL_KNN), the instantiated matcher (either brute force or FLANN) is then passed to the called (knnMatch). With k=2, this instructs the function to find the 2 best neighbouring matches. The output is a (cv::DMatch) type variable called (knn_matches) where these matches are stored.

```
matcher->knnMatch(descSource, descRef, knn_matches, 2);
```

A for loop is then used for each iteration of match in the (knn_matches) vector to filter matches using the descriptor ratio test. A variable of (Double) type called (minDescDistRatio) is set to 0.8 and is used for the filtering. Any points that are considered outside of this threshold are discarded and any matches inside this threshold are taken forward as positive matches for further consideration.

Performance

MP.7 Performance Evaluation 1

Count the number of keypoints on the preceding vehicle for all 10 images and take note of the distribution of their neighborhood size. Do this for all the detectors you have implemented.

For each detector and each of the 9 available images, the table on the following page is the number of resultant keypoints of the preceding vehicle with respective neighbourhood size.

Image Number	Detector Name	No. of Keypoints on preceding vehicle	Neighbourhood Size
1	SHITOMASI	118	4
2	SHITOMASI	123	4
3	SHITOMASI	120	4
4	SHITOMASI	120	4
5	SHITOMASI	113	4
6	SHITOMASI	114	4
7	SHITOMASI	123	4
8	SHITOMASI	111	4
9	SHITOMASI	112	4
1	HARRIS	14	6
2	HARRIS	18	6
3	HARRIS	21	6
4	HARRIS	26	6
5	HARRIS	43	6
6	HARRIS	18	6
7	HARRIS	31	6
8	HARRIS	26	6
9	HARRIS	34	6
1	FAST	152	7
2	FAST	150	7
3	FAST	155	7
4	FAST	149	7
5	FAST	149	7
6	FAST	156	7
7	FAST	150	7
8	FAST	138	7
9	FAST	143	7
1	BRISK	282	72
2	BRISK	282	72
3	BRISK	277	72
4	BRISK	297	72
5	BRISK	279	72
6	BRISK	289	72
7	BRISK	272	72
8	BRISK	266	72
9	BRISK	254	72
1	ORB	102	111.079
2	ORB	106	111.079
3	ORB	113	111.079
4	ORB	109	111.079
5	ORB	125	111.079
6	ORB	130	111.079
7	ORB	129	111.079
8	ORB	127	111.079
9	ORB	128	111.079
1	AKAZE	157	16.1452
2	AKAZE	161	16.1452
3	AKAZE	155	16.1452
4	AKAZE	163	16.1452
5	AKAZE	164	16.1452
6	AKAZE	173	16.1452
7	AKAZE	175	16.1452
8	AKAZE	177	16.1452
9	AKAZE	179	16.1452
1	SIFT	132	2.35782
2	SIFT	124	2.35782
3	SIFT	138	2.35782
4	SIFT	134	2.35782
5	SIFT	140	2.35782
6	SIFT	137	2.35782
7	SIFT	148	2.35782
8	SIFT	159	2.35782
9	SIFT	137	2.35782

MP.8 Performance Evaluation 2

Count the number of matched keypoints for all 10 images using all possible combinations of detectors and descriptors. In the matching step, the BF approach is used with the descriptor distance ratio set to 0.8.

For each detector and descriptor variation possible under the consideration the 9 available array of images, the table on the following pages is the number of resultant keypoints of the preceding vehicle.

These results follow the brute force (MAT_BF) matcher method with the minimum descriptor distance ratio (minDescDistRatio) set to 0.8.

MP.9 Performance Evaluation 3

Log the time it takes for keypoint detection and descriptor extraction. The results must be entered into a spreadsheet and based on this data, the TOP3 detector / descriptor combinations must be recommended as the best choice for our purpose of detecting keypoints on vehicles.

As with MP.8, the same resultant table below should be referenced, the timings taken for detection and description computation respectively are shown for all detector and descriptor variations possible for the 9 image array.

My top 3 detector and descriptor combinations chosen are:-

- 1 – Detector(FAST), Descriptor(BRIEF)
- 2 – Detector(FAST), Descriptor(BRISK)
- 3 – Detector(FAST), Descriptor(ORB)

My choice for detector being (FAST) is due to the consistent time taken being < 1ms, with keypoint matches consistently above 60. In comparison to other detectors where variations in time taken ranging from 1.8ms to 200ms and keypoint matches ranging from 9 to 203. Most other detectors with high numbers of keypoints matches have a high correlation with time taken, for example SIFT.

My choice for descriptors is again due to the consistent time taken being at a minimum 3ms faster than all other descriptor types. In the top 10 positions for minimum time taken BRIEF shows up 5 times, BRISK 4 times and ORB once. With the variation between all 10 being only 0.189ms.

The fastest FAST/BRIEF combination exhibited a detection time of (0.810132) and a description time of (0.429209) with the number of matched keypoints being 107. Whilst this wasn't the fastest detection and description time combination, the number of keypoints exceeds the next closest by 20 points when also considering the time taken. As our target is to accurately and consistently detect and track the preceding vehicle. The number of keypoints should be considered with importance to maximise the accuracy of the TTC calculation. With this established its then necessary to balance a method that delivers timings close to real time due to the application requiring a real time response.

A CSV file with all results can be found in results folder in the root project directory:-

SFND_2D_Feature_Tracking-master-/results/ Camera Based 2D Feature Tracking - Mid-term project - Chris Holmes - MP7_8_9 Results.xlsx

Image Number	Detector Name	Descriptor Name	Number of matched keypoints	Detection Time (ms)	Description Time (ms)
1	SHITOMASI	BRISK	84	10.6146	1.48467
2	SHITOMASI	BRISK	80	14.6251	1.57598
3	SHITOMASI	BRISK	73	10.3995	1.43193
4	SHITOMASI	BRISK	77	9.57576	1.05954
5	SHITOMASI	BRISK	74	7.55957	1.12861
6	SHITOMASI	BRISK	70	18.0541	1.03618
7	SHITOMASI	BRISK	79	7.83425	1.06085
8	SHITOMASI	BRISK	81	12.1282	0.974716
9	SHITOMASI	BRISK	72	10.6239	1.4152
1	SHITOMASI	BRIEF	96	10.3994	0.442755
2	SHITOMASI	BRIEF	93	26.2236	1.38109
3	SHITOMASI	BRIEF	92	10.7263	0.812701
4	SHITOMASI	BRIEF	89	28.5357	1.35425
5	SHITOMASI	BRIEF	92	10.4042	0.826384
6	SHITOMASI	BRIEF	93	10.3601	0.929265
7	SHITOMASI	BRIEF	85	14.6635	0.822054
8	SHITOMASI	BRIEF	91	15.0815	0.836312
9	SHITOMASI	BRIEF	85	10.136	0.785321
1	SHITOMASI	ORB	86	13.0292	0.768034
2	SHITOMASI	ORB	84	10.6318	0.677963
3	SHITOMASI	ORB	87	14.7691	0.144426
4	SHITOMASI	ORB	91	10.0127	0.668367
5	SHITOMASI	ORB	87	10.2386	0.726097
6	SHITOMASI	ORB	76	10.0876	0.662108
7	SHITOMASI	ORB	81	10.4274	0.71879
8	SHITOMASI	ORB	88	14.6107	0.701962
9	SHITOMASI	ORB	88	9.43086	0.677713
1	SHITOMASI	FREAK	66	11.8866	27.4224
2	SHITOMASI	FREAK	66	9.96683	27.1257
3	SHITOMASI	FREAK	64	8.09261	26.9747
4	SHITOMASI	FREAK	63	7.54342	27.637
5	SHITOMASI	FREAK	62	12.574	27.779
6	SHITOMASI	FREAK	64	8.36919	28.0374
7	SHITOMASI	FREAK	61	7.96683	28.2477
8	SHITOMASI	FREAK	65	7.7073	27.7641
9	SHITOMASI	FREAK	63	7.70595	29.8493
1	SHITOMASI	AKAZE	AKAZE DETECTOR ONLY		
2	SHITOMASI	AKAZE	AKAZE DETECTOR ONLY		
3	SHITOMASI	AKAZE	AKAZE DETECTOR ONLY		
4	SHITOMASI	AKAZE	AKAZE DETECTOR ONLY		
5	SHITOMASI	AKAZE	AKAZE DETECTOR ONLY		
6	SHITOMASI	AKAZE	AKAZE DETECTOR ONLY		
7	SHITOMASI	AKAZE	AKAZE DETECTOR ONLY		
8	SHITOMASI	AKAZE	AKAZE DETECTOR ONLY		
9	SHITOMASI	AKAZE	AKAZE DETECTOR ONLY		
1	SHITOMASI	SIFT	112	8.35159	11.4756
2	SHITOMASI	SIFT	109	10.9746	11.7621
3	SHITOMASI	SIFT	104	13.6699	10.9685
4	SHITOMASI	SIFT	103	7.724	8.00867
5	SHITOMASI	SIFT	99	7.83194	8.01112
6	SHITOMASI	SIFT	101	7.67779	7.65197
7	SHITOMASI	SIFT	96	8.66958	7.99118
8	SHITOMASI	SIFT	106	7.74826	9.57518
9	SHITOMASI	SIFT	97	7.69153	7.88452
1	HARRIS	BRISK	11	11.7355	0.283911
2	HARRIS	BRISK	9	10.1153	0.304755
3	HARRIS	BRISK	10	8.26248	0.332856
4	HARRIS	BRISK	11	12.2634	0.365635
5	HARRIS	BRISK	16	16.7142	0.487819
6	HARRIS	BRISK	14	7.44458	0.307754
7	HARRIS	BRISK	12	9.31704	0.400533
8	HARRIS	BRISK	21	8.13152	0.406479
9	HARRIS	BRISK	17	11.0106	0.428091
1	HARRIS	BRIEF	12	10.4295	0.147346
2	HARRIS	BRIEF	12	10.0602	0.144972
3	HARRIS	BRIEF	14	8.10512	0.598747
4	HARRIS	BRIEF	17	10.2889	0.65219
5	HARRIS	BRIEF	17	17.772	0.714542
6	HARRIS	BRIEF	16	14.1229	0.653261
7	HARRIS	BRIEF	12	11.1058	0.659409
8	HARRIS	BRIEF	20	10.6991	0.180136
9	HARRIS	BRIEF	21	11.6562	0.671966
1	HARRIS	ORB	11	11.8612	0.595843
2	HARRIS	ORB	11	11.2017	0.663667
3	HARRIS	ORB	14	12.7495	0.607037
4	HARRIS	ORB	17	10.1603	0.762234

5	HARRIS	ORB	19	15.7834	0.663059
6	HARRIS	ORB	19	7.42563	0.748412
7	HARRIS	ORB	13	9.11381	0.702721
8	HARRIS	ORB	21	12.8098	0.601752
9	HARRIS	ORB	20	16.4293	0.749771
1	HARRIS	FREAK	11	11.2128	26.5586
2	HARRIS	FREAK	9	9.09957	27.2037
3	HARRIS	FREAK	13	8.06505	27.3148
4	HARRIS	FREAK	14	8.50702	27.4895
5	HARRIS	FREAK	13	16.0707	26.9198
6	HARRIS	FREAK	18	7.73141	27.176
7	HARRIS	FREAK	10	9.62944	28.501
8	HARRIS	FREAK	17	8.29489	26.3783
9	HARRIS	FREAK	18	27.6625	29.5114
1	HARRIS	AKAZE	AKAZE DETECTOR ONLY		
2	HARRIS	AKAZE	AKAZE DETECTOR ONLY		
3	HARRIS	AKAZE	AKAZE DETECTOR ONLY		
4	HARRIS	AKAZE	AKAZE DETECTOR ONLY		
5	HARRIS	AKAZE	AKAZE DETECTOR ONLY		
6	HARRIS	AKAZE	AKAZE DETECTOR ONLY		
7	HARRIS	AKAZE	AKAZE DETECTOR ONLY		
8	HARRIS	AKAZE	AKAZE DETECTOR ONLY		
9	HARRIS	AKAZE	AKAZE DETECTOR ONLY		
1	HARRIS	SIFT	14	8.15746	10.1825
2	HARRIS	SIFT	11	9.17969	7.72386
3	HARRIS	SIFT	16	7.74376	7.19132
4	HARRIS	SIFT	19	11.85	7.43046
5	HARRIS	SIFT	22	16.0043	7.41266
6	HARRIS	SIFT	22	7.39724	7.35759
7	HARRIS	SIFT	13	13.5217	7.31115
8	HARRIS	SIFT	24	12.269	7.51149
9	HARRIS	SIFT	22	14.878	7.82184
1	FAST	BRISK	80	0.850181	1.67321
2	FAST	BRISK	91	3.34967	1.74055
3	FAST	BRISK	85	0.812284	1.52569
4	FAST	BRISK	94	0.808348	1.29577
5	FAST	BRISK	71	0.804647	1.25703
6	FAST	BRISK	83	0.807662	1.27477
7	FAST	BRISK	92	0.802393	1.25887
8	FAST	BRISK	86	0.815203	1.148
9	FAST	BRISK	94	0.812284	1.18253
1	FAST	BRIEF	88	0.587875	1.32291
2	FAST	BRIEF	102	0.816636	0.907411
3	FAST	BRIEF	88	0.808233	0.592179
4	FAST	BRIEF	102	0.802878	0.565507
5	FAST	BRIEF	94	3.46394	0.588891
6	FAST	BRIEF	98	0.802967	0.576116
7	FAST	BRIEF	112	3.79209	0.45914
8	FAST	BRIEF	107	0.810132	0.429209
9	FAST	BRIEF	92	0.813631	0.570418
1	FAST	ORB	96	0.916686	1.41858
2	FAST	ORB	102	0.565914	0.763725
3	FAST	ORB	87	0.809207	0.77223
4	FAST	ORB	94	0.805804	0.715612
5	FAST	ORB	87	3.91628	0.72989
6	FAST	ORB	100	3.46538	0.755828
7	FAST	ORB	101	0.801343	0.69989
8	FAST	ORB	96	0.813618	0.741928
9	FAST	ORB	99	3.96319	0.72533
1	FAST	FREAK	64	0.874496	30.2079
2	FAST	FREAK	80	0.581345	29.3177
3	FAST	FREAK	65	0.808794	27.894
4	FAST	FREAK	79	0.805594	26.7128
5	FAST	FREAK	61	0.857892	27.0974
6	FAST	FREAK	76	0.806846	28.9905
7	FAST	FREAK	83	1.79266	27.9437
8	FAST	FREAK	77	0.830148	27.6232
9	FAST	FREAK	82	3.7561	27.7886
1	FAST	AKAZE	AKAZE DETECTOR ONLY		
2	FAST	AKAZE	AKAZE DETECTOR ONLY		
3	FAST	AKAZE	AKAZE DETECTOR ONLY		
4	FAST	AKAZE	AKAZE DETECTOR ONLY		
5	FAST	AKAZE	AKAZE DETECTOR ONLY		
6	FAST	AKAZE	AKAZE DETECTOR ONLY		
7	FAST	AKAZE	AKAZE DETECTOR ONLY		
8	FAST	AKAZE	AKAZE DETECTOR ONLY		
9	FAST	AKAZE	AKAZE DETECTOR ONLY		
1	FAST	SIFT	118	0.919469	11.896

2	FAST	SIFT	123	3.92241	10.268
3	FAST	SIFT	110	0.80484	10.7829
4	FAST	SIFT	119	0.807533	10.514
5	FAST	SIFT	114	0.805418	9.12183
6	FAST	SIFT	119	3.82393	19.9078
7	FAST	SIFT	123	0.799815	8.93906
8	FAST	SIFT	117	0.808879	8.91129
9	FAST	SIFT	103	0.811942	8.94623
1	BRISK	BRISK	138	212.797	2.24644
2	BRISK	BRISK	144	209.795	2.23542
3	BRISK	BRISK	133	213.142	2.20698
4	BRISK	BRISK	144	214.8	2.41641
5	BRISK	BRISK	139	216.86	2.22958
6	BRISK	BRISK	155	213.415	2.30296
7	BRISK	BRISK	137	213.357	2.24955
8	BRISK	BRISK	150	211.336	2.23832
9	BRISK	BRISK	158	217.82	2.16143
1	BRISK	BRIEF	138	217.501	0.741463
2	BRISK	BRIEF	166	212.769	1.40944
3	BRISK	BRIEF	129	212.353	0.812571
4	BRISK	BRIEF	141	213.978	1.10354
5	BRISK	BRIEF	148	211.565	0.870976
6	BRISK	BRIEF	155	212.415	0.794877
7	BRISK	BRIEF	158	207.826	0.980568
8	BRISK	BRIEF	161	215.701	0.746653
9	BRISK	BRIEF	148	212.78	0.739914
1	BRISK	ORB	94	211.137	4.43103
2	BRISK	ORB	107	215.516	4.36541
3	BRISK	ORB	88	213.699	3.9734
4	BRISK	ORB	97	216.347	3.35789
5	BRISK	ORB	85	215.262	3.48744
6	BRISK	ORB	114	21.087	3.53303
7	BRISK	ORB	112	210.591	1.3035
8	BRISK	ORB	114	211.899	3.58739
9	BRISK	ORB	122	218.096	3.7612
1	BRISK	FREAK	114	223.037	30.1326
2	BRISK	FREAK	121	215.551	30.3784
3	BRISK	FREAK	113	216.043	28.7622
4	BRISK	FREAK	118	215.982	28.4592
5	BRISK	FREAK	203	213.018	27.7449
6	BRISK	FREAK	129	217.821	28.1343
7	BRISK	FREAK	135	211.812	27.8901
8	BRISK	FREAK	129	212.425	29.7067
9	BRISK	FREAK	131	213.979	27.411
1	BRISK	AKAZE	AKAZE DETECTOR ONLY		
2	BRISK	AKAZE	AKAZE DETECTOR ONLY		
3	BRISK	AKAZE	AKAZE DETECTOR ONLY		
4	BRISK	AKAZE	AKAZE DETECTOR ONLY		
5	BRISK	AKAZE	AKAZE DETECTOR ONLY		
6	BRISK	AKAZE	AKAZE DETECTOR ONLY		
7	BRISK	AKAZE	AKAZE DETECTOR ONLY		
8	BRISK	AKAZE	AKAZE DETECTOR ONLY		
9	BRISK	AKAZE	AKAZE DETECTOR ONLY		
1	BRISK	SIFT	182	209.463	14.8813
2	BRISK	SIFT	193	212.134	17.9454
3	BRISK	SIFT	169	213.66	18.1821
4	BRISK	SIFT	183	213.746	17.7909
5	BRISK	SIFT	171	213.613	14.9007
6	BRISK	SIFT	195	223.383	15.6104
7	BRISK	SIFT	194	211.819	14.9218
8	BRISK	SIFT	176	216.686	15.6658
9	BRISK	SIFT	183	212.827	14.308
1	ORB	BRISK	60	6.33516	0.933538
2	ORB	BRISK	65	9.41149	0.941125
3	ORB	BRISK	65	4.97267	1.01063
4	ORB	BRISK	76	4.85832	0.974233
5	ORB	BRISK	72	9.90616	1.10095
6	ORB	BRISK	83	5.08379	1.12777
7	ORB	BRISK	83	5.04191	1.22003
8	ORB	BRISK	73	5.8867	1.22574
9	ORB	BRISK	72	5.47493	1.1066
1	ORB	BRIEF	37	5.57879	0.314383
2	ORB	BRIEF	38	5.48387	0.326798
3	ORB	BRIEF	37	9.00037	0.361691
4	ORB	BRIEF	53	5.153	0.383235
5	ORB	BRIEF	42	5.08314	0.401183
6	ORB	BRIEF	64	5.05062	0.408883
7	ORB	BRIEF	58	9.40914	0.643216

8	ORB	BRIEF	62	5.0151	0.435306
9	ORB	BRIEF	59	5.0168	0.44572
1	ORB	ORB	40	5.54604	3.60117
2	ORB	ORB	57	5.48026	3.53091
3	ORB	ORB	49	5.58472	3.68902
4	ORB	ORB	54	5.30616	4.3966
5	ORB	ORB	57	4.91443	3.49654
6	ORB	ORB	68	4.98238	3.65071
7	ORB	ORB	71	5.42699	3.32483
8	ORB	ORB	62	9.53212	3.55561
9	ORB	ORB	72	5.11674	4.13298
1	ORB	FREAK	39	5.6615	28.0309
2	ORB	FREAK	33	8.82305	28.2947
3	ORB	FREAK	37	4.98501	27.1141
4	ORB	FREAK	40	5.08717	27.8646
5	ORB	FREAK	33	9.15411	27.9377
6	ORB	FREAK	40	5.16931	27.4063
7	ORB	FREAK	41	4.9494	26.2664
8	ORB	FREAK	39	7.42487	27.0874
9	ORB	FREAK	44	5.08369	26.6638
1	ORB	AKAZE	AKAZE DETECTOR ONLY		
2	ORB	AKAZE	AKAZE DETECTOR ONLY		
3	ORB	AKAZE	AKAZE DETECTOR ONLY		
4	ORB	AKAZE	AKAZE DETECTOR ONLY		
5	ORB	AKAZE	AKAZE DETECTOR ONLY		
6	ORB	AKAZE	AKAZE DETECTOR ONLY		
7	ORB	AKAZE	AKAZE DETECTOR ONLY		
8	ORB	AKAZE	AKAZE DETECTOR ONLY		
9	ORB	AKAZE	AKAZE DETECTOR ONLY		
1	ORB	SIFT	67	5.14507	17.3177
2	ORB	SIFT	79	5.20088	16.3582
3	ORB	SIFT	78	5.13727	17.4147
4	ORB	SIFT	79	4.94949	16.3072
5	ORB	SIFT	82	8.79609	16.2553
6	ORB	SIFT	95	9.63085	18.0393
7	ORB	SIFT	95	8.86575	20.417
8	ORB	SIFT	94	9.42087	20.7202
9	ORB	SIFT	94	5.0557	19.9204
1	AKAZE	BRISK	126	38.6191	1.28892
2	AKAZE	BRISK	112	46.8254	1.31887
3	AKAZE	BRISK	121	42.0875	1.33687
4	AKAZE	BRISK	117	42.1096	1.34647
5	AKAZE	BRISK	114	42.5457	1.38782
6	AKAZE	BRISK	119	45.2503	1.45901
7	AKAZE	BRISK	134	43.7245	1.41714
8	AKAZE	BRISK	140	46.3934	1.42907
9	AKAZE	BRISK	127	40.5905	1.51241
1	AKAZE	BRIEF	108	38.8803	0.496096
2	AKAZE	BRIEF	116	41.8734	0.487903
3	AKAZE	BRIEF	110	42.2974	0.486645
4	AKAZE	BRIEF	109	40.7625	0.513132
5	AKAZE	BRIEF	116	39.1067	0.509107
6	AKAZE	BRIEF	129	34.3945	0.525971
7	AKAZE	BRIEF	133	40.7909	0.531596
8	AKAZE	BRIEF	135	35.3616	0.538762
9	AKAZE	BRIEF	131	38.4601	0.576297
1	AKAZE	ORB	102	41.2922	2.37586
2	AKAZE	ORB	96	39.2419	2.71818
3	AKAZE	ORB	96	42.1979	2.40701
4	AKAZE	ORB	84	43.3145	2.4182
5	AKAZE	ORB	90	39.2515	2.3967
6	AKAZE	ORB	116	38.2929	2.42624
7	AKAZE	ORB	103	38.9726	2.84598
8	AKAZE	ORB	113	42.0892	2.80604
9	AKAZE	ORB	118	43.5489	2.72068
1	AKAZE	FREAK	103	38.6738	28.6237
2	AKAZE	FREAK	105	38.7226	27.5128
3	AKAZE	FREAK	93	33.6809	28.5962
4	AKAZE	FREAK	99	36.9601	27.2238
5	AKAZE	FREAK	97	32.3856	27.089
6	AKAZE	FREAK	115	33.4737	28.0373
7	AKAZE	FREAK	126	30.9871	27.086
8	AKAZE	FREAK	118	36.3162	27.214
9	AKAZE	FREAK	117	42.1814	29.2261
1	AKAZE	AKAZE	128	43.1631	32.3484
2	AKAZE	AKAZE	128	41.687	35.4047
3	AKAZE	AKAZE	125	39.3578	34.5493
4	AKAZE	AKAZE	117	45.5181	31.1786

5	AKAZE	AKAZE	121	38.8456	34.48
6	AKAZE	AKAZE	132	37.1915	37.1915
7	AKAZE	AKAZE	137	45.8882	34.9083
8	AKAZE	AKAZE	140	42.7672	38.9629
9	AKAZE	AKAZE	144	38.7955	34.0101
1	AKAZE	SIFT	134	38.4855	12.4091
2	AKAZE	SIFT	134	41.0002	14.7304
3	AKAZE	SIFT	130	45.3373	13.4999
4	AKAZE	SIFT	136	39.2837	14.4046
5	AKAZE	SIFT	137	38.006	10.5398
6	AKAZE	SIFT	147	40.4331	11.7292
7	AKAZE	SIFT	147	39.2171	13.8445
8	AKAZE	SIFT	154	38.3759	12.1161
9	AKAZE	SIFT	151	41.2647	13.0903
1	SIFT	BRISK	57	54.4008	1.08088
2	SIFT	BRISK	63	38.4389	1.05385
3	SIFT	BRISK	59	40.487	1.13909
4	SIFT	BRISK	62	43.5718	1.20749
5	SIFT	BRISK	55	40.3461	1.13907
6	SIFT	BRISK	52	40.5314	1.19875
7	SIFT	BRISK	54	42.0366	1.20641
8	SIFT	BRISK	63	40.1085	1.28557
9	SIFT	BRISK	73	48.5458	1.12873
1	SIFT	BRIEF	63	53.6397	0.478225
2	SIFT	BRIEF	72	44.2619	0.454381
3	SIFT	BRIEF	65	40.7103	0.458913
4	SIFT	BRIEF	67	39.63	0.474287
5	SIFT	BRIEF	52	43.2356	0.47881
6	SIFT	BRIEF	57	39.6669	0.521658
7	SIFT	BRIEF	72	42.3281	0.480099
8	SIFT	BRIEF	67	39.3939	0.484358
9	SIFT	BRIEF	84	47.3239	0.517459
1	SIFT	ORB	NOT COMPATIBLE		
2	SIFT	ORB	NOT COMPATIBLE		
3	SIFT	ORB	NOT COMPATIBLE		
4	SIFT	ORB	NOT COMPATIBLE		
5	SIFT	ORB	NOT COMPATIBLE		
6	SIFT	ORB	NOT COMPATIBLE		
7	SIFT	ORB	NOT COMPATIBLE		
8	SIFT	ORB	NOT COMPATIBLE		
9	SIFT	ORB	NOT COMPATIBLE		
1	SIFT	FREAK	59	54.5764	28.8607
2	SIFT	FREAK	63	41.6989	28.6973
3	SIFT	FREAK	55	38.2872	27.8335
4	SIFT	FREAK	65	38.9551	28.2883
5	SIFT	FREAK	51	39.4926	28.5389
6	SIFT	FREAK	50	43.5876	27.7344
7	SIFT	FREAK	47	44.752	28.0553
8	SIFT	FREAK	53	42.805	27.4287
9	SIFT	FREAK	65	42.0471	30.2738
1	SIFT	AKAZE	AKAZE DETECTOR ONLY		
2	SIFT	AKAZE	AKAZE DETECTOR ONLY		
3	SIFT	AKAZE	AKAZE DETECTOR ONLY		
4	SIFT	AKAZE	AKAZE DETECTOR ONLY		
5	SIFT	AKAZE	AKAZE DETECTOR ONLY		
6	SIFT	AKAZE	AKAZE DETECTOR ONLY		
7	SIFT	AKAZE	AKAZE DETECTOR ONLY		
8	SIFT	AKAZE	AKAZE DETECTOR ONLY		
9	SIFT	AKAZE	AKAZE DETECTOR ONLY		
1	SIFT	SIFT	82	40.4548	38.1691
2	SIFT	SIFT	81	43.5383	39.4963
3	SIFT	SIFT	86	40.8892	38.1301
4	SIFT	SIFT	94	39.0666	42.4418
5	SIFT	SIFT	90	43.0906	43.4629
6	SIFT	SIFT	81	44.2323	40.3504
7	SIFT	SIFT	82	40.1809	39.4129
8	SIFT	SIFT	102	39.4857	38.6441
9	SIFT	SIFT	104	37.8701	40.3224